BYUNG HYUN LEE

CONTACT INFORMATION

Affiliation: Department of Electrical and Computer Engineering, Seoul National University

Email Website Google Scholar LinkedIn

RESEARCH INTERESTS

My research focuses on *selective knowledge updates* for deep visual and foundation models — adding new knowledge or removing unwanted knowledge while preserving existing capabilities. I have studied this through continual learning and concept erasure approaches, essential for agentic/physical AI, vertical-AI, and visual generative AI. I am currently extending these ideas to multimodal LLMs, omnimodal models, and domain-specialized models via post-training with minimal forgetting, aiming at adaptive and reliable knowledge updates that efficiently scale to real-world settings.

Additionally, I'm interested in model acceleration, image restoration, and histopathology analysis.

EDUCATION

- **Seoul National University (SNU)** *Mar 2021 - Feb 2027 (expected)*
  Combined M.S./Ph.D Program in Electrical and Computer Engineering (ECE)
- **Ulsan National Institute of Science and Technology (UNIST)** *Mar 2015 - Aug 2018*
  B.S. in Electrical Engineering, GPA: 4.04/4.30

RESEARCH ADVISOR

- **Se Young Chun:** Professor, Department of Electrical and Computer Engineering (ECE), SNU

PUBLICATIONS (Top AI Conference Papers)

- Erasing Thousands of Concepts: Towards Scalable and Practical Concept Erasure for Text-to-Image Diffusion Models
  H. Seo*, **B. H. Lee***, J. Cho, S. Lim, S. Y. Chun
  Conference on Computer Vision and Pattern Recognition (CVPR), 2026
- Continual Multiple Instance Learning with Enhanced Localization for Histopathological Whole Slide Image Analysis
  B. H. Lee, W. Jeong, W. Han, K. Lee, S. Y. Chun
  International Conference on Computer Vision (ICCV), 2025
- Localized Concept Erasure for Text-to-Image Diffusion Models Using Training-Free Gated Low-Rank Adaptation
  B. H. Lee*, S. Lim*, S. Y. Chun (*co-first authors)
  Conference on Computer Vision and Pattern Recognition (CVPR), 2025
- Concept pinpoint eraser for text-to-image diffusion models via residual attention gate
  B. H. Lee*, S. Lim*, S. Lee, D. U. Kang, S. Y. Chun (*co-first authors)
  International Conference on Learning Representations (ICLR), 2025
- Doubly perturbed task free continual learning
  B. H. Lee, M. Oh, S. Y. Chun
  Proceedings of the AAAI Conference on Artificial Intelligence (AAAI, oral), 2024

- Online Continual Learning on Hierarchical Label Expansion
  B. H. Lee*, O. Jung*, J. Choi, S. Y. Chun (*co-first authors)
  International Conference on Computer Vision (ICCV), 2023
- All-in-one image restoration for unknown degradations using adaptive discriminative filters for specific degradations
  D. Park, **B. H. Lee**, S. Y. Chun
  Conference on Computer Vision and Pattern Recognition (CVPR), 2023

PUBLICATIONS (Journal Papers)

- Continual Test-Time Adaptation for Robust Remote Photoplethysmography Estimation
  H. Lee, H. Lee, **B. H. Lee**, S. Y. Chun
  IEEE Access, 2025
- Towards accelerating model parallelism in distributed deep learning systems
  H. Choi*, **B. H. Lee***, S. Y. Chun, J. Lee (*co-first authors)
  PLOS One, 2023

PUBLICATIONS (Short, Workshop, or Domestic Papers)

- Unlearning the Unpromptable: Prompt-free Instance Unlearning in Diffusion Models
  K. R. Lee*, K. H. Lee*, S. Hong, **B. H. Lee** , S. Y. Chun
  CVPR Workshop on Machine Unlearning for Computer Vision, 2026
- Selective Concept Erasing for Safe Diffusion Models
  B. H. Lee, S. Lim, S. Y. Chun
  Korea Signal Processing Conference (Best Poster Presentation Award), 2024
- Expert classifier ensemble based post-processing correction for unbiased scene graph generation
  S. Lee, **B. H. Lee**, S. Y. Chun
  Workshop on Image Processing and Image Understanding (IPIU), 2024
- Efficient and accurate quantized image super-resolution on mobile npus, mobile ai & aim 2022 challenge: report
  A. Ignatov et al. (including **B. H. Lee**)
  European Conference on Computer Vision Workshops (ECCVW), 2022
- Efficient single-image depth estimation on mobile devices, mobile AI & AIM 2022 challenge: report
  A. Ignatov et al. (including **B. H. Lee**)
  European Conference on Computer Vision Workshops (ECCVW), 2022
- Uncertainty-based dual domain low-dose X-ray CT reconstruction
  S. Lee, D. U. Kang, **B. H. Lee**, S. Y. Chun
  Korean Signal Processing Conference, 2022
- Empirically Accelerating Scaled Gradient Projection Using Deep Neural Network for Inverse Problems in Image Processing
  B. H. Lee, S. Y. Chun
  International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2021

TECHNICAL REPORTS / MANUSCRIPTS UNDER REVIEW

- Strong Helps Weak: Directional Cross-Modal Alignment Transfer in Multi-modal LLMs
  Under review, 2026
- Speculative Encoding for Efficient Gigapixel Whole Slide Image Analysis
  Under review, 2026
- HYPERION: Joint-Hierarchy Hyperbolic Fine-Tuning for Histopathology Slide Representations
  Under review, 2026

- RePath: Zero-Shot Tumor Segmentation in Gigapixel WSIs via Differential Intra-Slide Patch Affinity
  Under review, 2026
- HyperCLOVA X 8B Omni
  HyperCLOVA X Team in NAVER Cloud (including **B. H. Lee**)
  2026
- HyperCLOVA X 32B Think
  HyperCLOVA X Team in NAVER Cloud (including **B. H. Lee**)
  2026
- Geometrical Properties of Text Token Embeddings for Strong Semantic Binding in Text-to-Image Generation
  H. Seo*, J. Bang*, H. Lee*, J. Lee, **B. H. Lee** , S. Y. Chun
  arXiv, 2025

PATENTS

- **Training-free Industrial Defect Segmentation.**
  S. Y. Chun, D. U. Kang, H. Kim, J. H. Jang, **B. H. Lee** & Hoigi Seo
  Korean Patent, Filed. 2025.
- **All-in-one image quality improvement model providing method performing image quality restoration for multiple image quality degradation factors**
  S. Y. Chun, D. Park, **B. H. Lee**
  U.S. Patent, Registered, 2026 (Filed, 2023)
- **Method for providing all-in-one image quality improvement model that performs image quality restoration for multiple image quality inhibitors**
  S. Y. Chun, D. Park, **B. H. Lee**
  Korea Patent, Filed, 2023

AWARDS AND HONORS

- **Compute Transparency Champion Award** *2026*
  CVPR 2026 (for "Erasing Thousands of Concepts")
- **Outstanding Reviewer (Top 5%)** *2026*
  Conference on Computer Vision and Pattern Recognition (CVPR)
- **Apple Scholars in AI/ML PhD Fellowship Program** *2025*
  Institutional Nominee
- **Qualcomm Innovations Fellowship Korea** *2025*
  Best paper finalist
- **2nd Place in Report Generation in Pathology using Pan-Asia Gigapixel WSIs** *2025*
  Challenge by International Conference on MICCAI (Served as the team leader, 26 teams participated)
- **Yulchon AI Star Scholarships** (₩8,000,000) *2025*
  Youlchon Foundation & SNU AI Institute

MEDIA COVERAGE

- **Popular Science Korea** *Jun 2026*
  Featured article on "Erasing Thousands of Concepts" (CVPR 2026), which erases 2,072 concepts from a text-to-image diffusion model

EXPERIENCES

- **National AI Foundation Model Project in South Korea**, NAVER Cloud *Oct 2025 - Feb 2026*
  Visiting Student Researcher
- **Biomedical Medical Image Processing Lab**, UNIST *May 2020 - Feb 2021*
  Researcher (Advisor: Prof. Se Young Chun)
- **Biomedical Medical Image Processing Lab**, UNIST *July 2017 - Aug 2018*
  Student Intership (Advisor: Prof. Se Young Chun)

PRESENTATIONS

- **Continual Learning and Its Applications in Magnetic Resonance Imaging**
  Advanced neuroimaging and AI workshop, SNU, 2024

ACADEMIC SERVICES

- **Reviewer**: NeurIPS, CVPR, ECCV, IEEE Transactions on Image Processing (TIP), Pattern Recognition

PROJECTS

- **AI for Human-Machine Teaming** *Mar 2025 - Present*
  United States Air Force & IITP, Republic of Korea
- **Industrial Defect Segmentation** *Mar 2024 - Mar 2025*
  Samsung Device Solution (DS)
- **AI System for Gigapixel Histopathological WSI Analysis** *Aug 2023 - Dec 2024*
  Ministry of Health and Welfare, Republic of Korea
- **Relations and Affordance Aware Robot Grasping** *May 2022 - May 2023*
  Samsung Research, Republic of Korea

EXTRACURRICULAR EXPERIENCES

- **Military Service, Republic of Korea Army** *Sep 2018 - Apr 2020*
  Discharged as Sergeant
- **Pinocchio (Robot Club)**, UNIST *Mar 2015 - Aug 2018*
  Education Director